ML Based Student Advisor

Project Report

Course Title: Artificial Intelligence lab
Course Code: CSE 422

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Date of Submission: December 13, 2025

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Motivation

The inspiration for this project emerged from a simple yet profound observation: behind every student grade is a unique human story. As class sizes grow and educational resources become increasingly stretched, many students find themselves navigating academic challenges without the personalized support they need to thrive. Teachers, though deeply committed to their students' success, often lack the time and tools to provide individualized attention to each learner.

We recognized that student performance is influenced by a complex interplay of factors extending far beyond the classroom. Academic struggles may stem from inadequate study habits, health concerns, family dynamics, social pressures, or simply a lack of awareness about effective learning strategies. Traditional educational approaches often address these issues only after problems become apparent, when early intervention could have made a significant difference.

This project was born from the belief that technology, when thoughtfully applied, can bridge the gap between individual student needs and the constraints of modern education systems. By combining data analysis with artificial intelligence, we aimed to create a system that understands the whole student, not just their test scores. Our motivation was to empower students with self-awareness, support educators with insights, and ultimately contribute to more equitable and effective educational outcomes for all learners.

Objectives

The primary goal of this project was to develop an intelligent system that could provide personalized academic guidance to students based on their unique circumstances. We established several specific objectives to guide our work.

First, we aimed to build accurate predictive models capable of estimating final student grades based on various academic and personal factors. Second, we

sought to create a reliable risk classification system that could identify students who might need additional support, categorizing them as high, medium, or low risk based on their predicted performance.

Third, we wanted to develop an AI-powered recommendation engine that could generate personalized, actionable advice for each student. These recommendations needed to be specific, practical, and tailored to the individual's situation, addressing not only study strategies but also lifestyle factors that impact learning.

Fourth, we intended to create a user-friendly interface that would allow educators, students, or parents to input student information and receive comprehensive analysis. Fifth, we planned to implement an automated reporting system that would generate professional PDF and text documents summarizing the analysis and recommendations.

Finally, we aimed to validate our system using real student data, ensuring its predictions were accurate and its recommendations were genuinely helpful. Throughout this process, we maintained a commitment to creating technology that served human needs, prioritizing interpretability, fairness, and practical utility over purely technical metrics.

Methodology

Data Collection

Our work began with data from real students, which provided the foundation for all subsequent analysis. We utilized the Student Performance Dataset from the UCI Machine Learning Repository, which contains information about secondary school students in Portugal. This dataset includes two separate files: one tracking mathematics performance and another tracking Portuguese language performance.

The combined dataset represented 1,044 individual students, each described by 33 different attributes. These attributes covered a comprehensive range of factors including demographic information such as age and gender, academic records including previous grades and number of past failures, study habits like

weekly study time, and lifestyle factors such as health status, alcohol consumption, and social activities.

We selected this dataset specifically because it represents real educational challenges rather than theoretical scenarios. The data captures the complex reality of student life, including both academic and non-academic factors that influence learning. By working with actual student records, we ensured our system would be grounded in the authentic experiences of learners.

Data Preprocessing

Before we could begin analysis, we needed to prepare the data for our models. This process involved several important steps to ensure the information was clean, consistent, and suitable for machine learning algorithms.

First, we examined the dataset for missing values and inconsistencies. Fortunately, the UCI dataset was well-maintained with few missing entries. For the small number of missing values we encountered, we implemented appropriate imputation strategies that preserved the integrity of the data.

Next, we needed to separate the various types of data for appropriate processing. Numerical data such as grades, age, and study time required different treatment than categorical data such as school type, family support availability, or internet access. We created separate processing pipelines for these different data types.

For numerical features, we applied standardization, which rescales the data to have a mean of zero and standard deviation of one. This ensures that all numerical features contribute equally to the model regardless of their original scale. For categorical features, we used one-hot encoding, which converts categorical variables into a format that can be provided to machine learning algorithms.

We also created our target variables during this stage. For regression analysis, our target was the final grade, a continuous value between 0 and 20. For classification, we created risk categories by grouping final grades: 0-8 as high risk, 9-12 as medium risk, and 13-20 as low risk. This categorization allowed us to address different student needs appropriately.

Finally, we split the data into training and testing sets, ensuring that our models would be evaluated on data they had never seen before. We used 80 percent of the data for training and reserved 20 percent for testing, implementing stratified sampling for the classification task to maintain the proportional representation of each risk category.

Training

With our data prepared, we began developing the machine learning models that would form the analytical core of our system. We selected Random Forest algorithms for both regression and classification tasks due to their robustness, interpretability, and strong performance with mixed data types.

The regression model was designed to predict final grades as continuous values. We trained this model using the prepared student data, with the final grade as the target variable. The model learned to recognize patterns and relationships between various student characteristics and their academic outcomes. We carefully tuned the model parameters to balance complexity with generalization, ensuring it would perform well on new, unseen student data.

Simultaneously, we trained a classification model to categorize students into risk levels. This model learned to identify the characteristics associated with each risk category, enabling it to classify new students appropriately. The classification task is particularly important for educational applications, as it helps prioritize interventions for students who need them most.

During training, we implemented cross-validation techniques to ensure our models were not overfitting to the training data. This process involves repeatedly training and testing the model on different subsets of the data, providing a more reliable estimate of how it will perform on new information.

We also conducted feature importance analysis to understand which factors most strongly influenced student performance. This analysis served dual purposes: it helped validate that our models were learning meaningful patterns, and it provided insights that could inform educational practice beyond the immediate predictions.

Alongside the machine learning components, we integrated the GroqCloud AI system for generating personalized recommendations. We developed carefully crafted prompts that framed the AI as an experienced educational psychologist, ensuring the recommendations would be evidence-based, compassionate, and practical. The AI system takes the model predictions along with detailed student profiles to generate customized advice covering immediate priorities, study strategies, lifestyle adjustments, resource utilization, long-term planning, and motivational support.

Results and Discussion

Our system achieved strong performance across multiple metrics, demonstrating its potential as a valuable educational tool. The regression model, designed to predict final grades, explained 81.2 percent of the variance in student performance. This high level of predictive accuracy suggests the model successfully identified meaningful patterns in the data. The model achieved a root mean squared error of 1.707 points on the 20-point grading scale, meaning its predictions were typically within about 1.7 points of the actual grades. The mean absolute error was 0.963 points, further confirming the model's reliability.

The classification model, tasked with identifying student risk levels, achieved 83.3 percent overall accuracy. More importantly, it demonstrated strong performance on the most critical categories: it correctly identified 77.4 percent of high-risk students who needed immediate attention, and 92.0 percent of low-risk students who were performing well. These results indicate the model can effectively prioritize support for students who need it most while accurately recognizing those who are succeeding.

The AI recommendation system successfully generated personalized advice for all test cases. The recommendations were consistently specific, actionable, and appropriately tailored to each student's predicted risk level and individual circumstances. The system maintained a response time of three to five seconds per analysis, making it practical for real-world use.

Several interesting patterns emerged from our analysis that have important implications for educational practice. We discovered that moderate study time,

typically five to ten hours per week, was associated with better outcomes than either minimal or excessive study time. This finding suggests that study quality and consistency may be more important than sheer quantity. We also found that weekend lifestyle factors, particularly social activities and alcohol consumption, showed stronger correlations with academic performance than weekday habits. This highlights the importance of balance and recovery in student success.

Family support emerged as a particularly strong predictor of student performance, correlating more strongly with academic success than extra paid classes or additional school resources. This finding emphasizes the crucial role families play in education and suggests that supporting family involvement could be a highly effective intervention strategy.

Health status showed a notable correlation with academic consistency, with students reporting better health generally demonstrating more stable performance across grading periods. This connection between physical wellbeing and academic performance warrants greater attention in educational planning and student support services.

Our system also revealed that internet access at home was significantly associated with better academic outcomes, likely reflecting both the practical benefits of online resources and the socioeconomic factors associated with internet availability. This finding has important implications for addressing educational inequality.

The feature importance analysis provided valuable insights into which factors most strongly influenced student performance. Previous academic performance, particularly second-period grades, emerged as the strongest predictor of final outcomes. Past failures and attendance records also showed significant importance, highlighting how academic history creates momentum that affects current performance. Study habits, family support, and lifestyle factors rounded out the most influential features, confirming that student success depends on a combination of academic and non-academic factors.

Throughout our work, we maintained a focus on creating technology that serves real human needs. The combination of accurate predictions with personalized, actionable recommendations represents a significant step toward more

responsive and effective educational support systems. By providing specific guidance rather than general advice, the system empowers students to take concrete steps toward improvement. By highlighting risk levels and influential factors, it helps educators prioritize their attention and tailor their interventions.

Conclusion

The Student Performance Analyzer project successfully developed an intelligent system that combines machine learning predictions with AI-generated recommendations to provide personalized academic guidance. Our work demonstrates that technology can play a valuable role in supporting student success when designed with careful attention to educational realities and human needs.

The system's strong predictive performance, with 81.2 percent variance explained for grade prediction and 83.3 percent accuracy for risk classification, confirms that machine learning models can identify meaningful patterns in student data. More importantly, the integration of these predictions with thoughtful AI recommendations creates a tool that goes beyond simple analytics to offer practical guidance students can use to improve their academic outcomes.

Several key insights emerged from our work that have broader implications for education. The importance of moderate, consistent study habits over excessive cramming suggests that educational messaging should emphasize quality and sustainability over quantity alone. The strong correlation between family support and academic success highlights the need for educational systems to engage families as partners in learning. The connection between lifestyle factors and academic performance reminds us that students are whole people whose learning is influenced by their overall wellbeing.

Looking forward, this project establishes a foundation that could be built upon in several valuable directions. Future work could expand the system to incorporate real-time data from learning management systems, allowing for more timely interventions as challenges emerge. Integration with school information systems could enable broader implementation across educational institutions.

Development of mobile applications could make the system more accessible to students and parents outside formal educational settings.
Perhaps most importantly, this project represents a shift in how we approach educational technology. Rather than replacing human educators, the system aims to amplify their ability to understand and support each student. By providing data-driven insights paired with personalized recommendations, it creates opportunities for more meaningful connections between educators and learners. In an educational landscape often dominated by standardized approaches and limited resources, tools like this one offer a pathway toward more personalized, responsive, and equitable learning experiences.

The ultimate success of any educational technology lies not in its technical specifications but in its ability to make a positive difference in students' lives. By focusing on practical utility, interpretable results, and actionable recommendations, the Student Performance Analyzer takes meaningful steps toward this goal. It represents our belief that every student deserves support tailored to their unique circumstances, and that technology, thoughtfully applied, can help make this vision a reality.